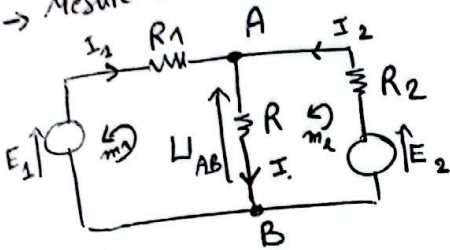


# Trouver les pratiques : Circuits électriques et électroniques

## TP1 : Utilisations des théorèmes généraux de l'électricité

### Partie théorique :

→ Mesure directe :



ii. loi de Kirchhoff :

→ loi des mailles :  $m_1 \quad U_{AB} + R_1 I_1 = E_1$   
 $m_2 \quad U_{AB} + R_2 I_2 = E_2$

→ loi des nœuds (A) :  $I = I_1 + I_2$

→ loi d'ohme (R) :  $I = \frac{U_{AB}}{R}$

$$\Rightarrow \begin{cases} I_1 = \frac{-U_{AB} + E_1}{R_1} \\ I_2 = \frac{-U_{AB} + E_2}{R_2} \\ \frac{U_{AB}}{R} = \frac{-U_{AB} + E_1}{R_1} + \frac{-U_{AB} + E_2}{R_2} \end{cases}$$

$$\Rightarrow U_{AB} \left( \frac{1}{R} + \frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{E_1}{R_1} + \frac{E_2}{R_2}$$

$$\Rightarrow U_{AB} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2}}{\frac{1}{R} + \frac{1}{R_1} + \frac{1}{R_2}} \times R_1 R_2$$

$$\Rightarrow U_{AB} = \frac{E_1 R_2 + E_2 R_1}{\frac{R_1 R_2}{R} + R_2 + R_1}$$

$$\Rightarrow U_{AB} = R \cdot \frac{E_1 R_2 + E_2 R_1}{R(R_1 + R_2) + R_1 R_2}$$

ii.  $I = \frac{U_{AB}}{R}$

$$\Rightarrow I = \frac{E_1 R_2 + E_2 R_1}{R(R_1 + R_2) + R_1 R_2}$$

$$\begin{aligned} I_1 &= \frac{E_1 - U_{AB}}{R_1} \\ &= \frac{E_1}{R_1} - \frac{1}{R_1} \cdot R \cdot \frac{E_1 R_2 + E_2 R_1}{R(R_1 + R_2) + R_1 R_2} \end{aligned}$$

$$\Rightarrow I_1 = \frac{E_1 (R(R_1 + R_2) + R_1 R_2) - R(E_1 R_2 + E_2 R_1)}{R_1 (R(R_1 + R_2) + R_1 R_2)}$$

$$= \frac{R_1 (E_1 (R + R_2) - E_2 R)}{R_1 (R(R_1 + R_2) + R_1 R_2)}$$

$$\Rightarrow I_1 = \frac{E_1 (R + R_2) - E_2 R}{R(R_1 + R_2) + R_1 R_2}$$

$$\begin{aligned} I_2 &= I - I_1 \\ &= \frac{E_1 R_2 + E_2 R_1 - E_1 (R + R_2) + E_2 R}{R(R_1 + R_2) + R_1 R_2} \end{aligned}$$

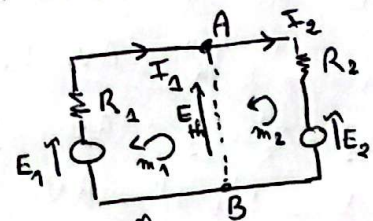
$$\Rightarrow I_2 = \frac{E_2 (R + R_1) - E_1 R}{R(R_1 + R_2) + R_1 R_2}$$

iii. A.N :  $U_{AB} = 4,36 \text{ V}$  ;  $I = 9,28 \text{ mA}$   
 $I_1 = 1,64 \text{ mA}$  ;  $I_2 = 7,64 \text{ mA}$

→ Théorème de Thévenin :

i. a  $E_{th}$  :

On débarrasse R



Méthode 1 : loi des mailles

$$\begin{aligned} m_1 \quad E_{th} &= E_1 - R_1 I \\ m_2 \quad E_{th} &= E_2 + R_2 I \\ I_1 &= I_2 = I \end{aligned} \Rightarrow \begin{cases} E_{th} = E_1 - R_1 I \\ I = \frac{E_{th} - E_2}{R_2} \end{cases}$$

$$\Rightarrow E_{th} = E_1 - R_1 \cdot \frac{(E_{th} - E_2)}{R_2}$$

$$\Rightarrow E_{th} \left( 1 + \frac{R_1}{R_2} \right) = E_1 + E_2 \frac{R_1}{R_2}$$

$$\Rightarrow E_{th} = \frac{R_2}{R_1 + R_2} (E_1 + E_2 \frac{R_1}{R_2})$$

$$\Rightarrow E_{th} = \frac{E_1 R_2 + E_2 R_1}{R_1 + R_2}$$

Méthode 2 :

Théorème de Millan :

$$E_{th} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

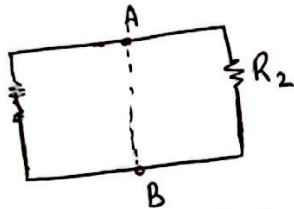
$$\Rightarrow E_{th} = \frac{E_1 R_2 + E_2 R_1}{R_1 + R_2}$$

\*  $R_{th}$

→ Débrancher R

→ court-circuiter les sources de tension

→ Déconnecter les sources de courant

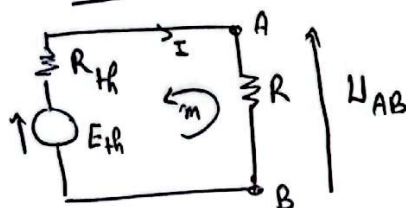


$R_1 \parallel R_2$

$$\Rightarrow R_{th} = R_{\text{ég}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$\Rightarrow R_{th} = \frac{R_1 R_2}{R_1 + R_2}$$

ii.



\* I : loi des mailles : m |  $E_{th} = U_{AB} + R_{th} I$

$$\Rightarrow E_{th} = I (R_{th} + R)$$

$$\Rightarrow I = \frac{E_{th}}{R + R_{th}}$$

\*  $U_{AB} = R I$  : loi de Ohm.

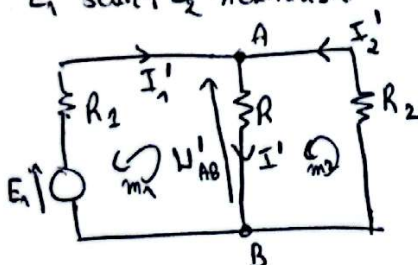
$$\Rightarrow U_{AB} = R \cdot \frac{E_{th}}{R + R_{th}}$$

iii. A.N :  $E_{th} = 9V$  ;  $R_{th} = 500\Omega$

$I = 9,28 \text{ mA}$  ;  $U_{AB} = 4,36V$

→ Théorème de superposition :

A  $E_1$  seule,  $E_2$  neutralisée



Lois de Kirchhoff :

$$\begin{cases} m_1 & E_1 = U_{AB}' + I_1' R_1 \\ m_2 & U_{AB}' + R_2 I_2' = 0 \\ (A) & I' = I_1' + I_2' = \frac{U_{AB}'}{R} \end{cases}$$

$$\Rightarrow \begin{cases} I_2' = \frac{E_1 - U_{AB}'}{R_1} \\ I_2' = -\frac{U_{AB}'}{R_2} \\ \frac{U_{AB}'}{R} = \frac{E_1 - U_{AB}'}{R_1} - \frac{U_{AB}'}{R_2} \end{cases}$$

$$\Rightarrow U_{AB}' \left( \frac{1}{R} + \frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{E_1}{R_1}$$

$$\Rightarrow U_{AB}' = \frac{E_1}{R_1 \cdot \frac{R_1 R_2 + R R_2 + R R_1}{R R_1 R_2}}$$

$$\Rightarrow U_{AB}' = \frac{E_1 R R_2}{R_1 R_2 + R (R_1 + R_2)}$$

$$ii. \quad I' = \frac{U_{AB}'}{R} \Rightarrow I' = \frac{E_1 R_2}{R_1 R_2 + R (R_1 + R_2)}$$

$$* I_2' = -\frac{U_{AB}'}{R_2}$$

$$\Rightarrow I_2' = \frac{-E_1 R}{R_1 R_2 + R (R_1 + R_2)}$$

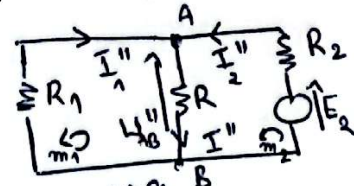
$$* I_1' = I' - I_2'$$

$$\Rightarrow I_1' = \frac{E_1 (R + R_2)}{R_1 R_2 + R (R_1 + R_2)}$$

iii. A.N :  $U_{AB}' = 1,45V$  ;  $I' = 3,09 \text{ mA}$

$I_1' = 4,55 \text{ mA}$  ;  $I_2' = -1,45 \text{ mA}$

B.  $E_2$  seule,  $E_1$  neutralisée :



i. Loi de Kirchhoff :

$$\begin{cases} m_1 & U_{AB}'' + R_1 I_1'' = 0 \\ m_2 & E_2 = U_{AB}'' + I_2'' R_2 \\ (A) & I'' = I_1'' + I_2'' = \frac{U_{AB}''}{R} \end{cases}$$

$$\Rightarrow \frac{U_{AB}''}{R} = -\frac{U_{AB}''}{R_1} + \frac{E_2 - U_{AB}''}{R_2}$$

$$U_{AB}'' = \frac{E_2}{R_2 \left( \frac{1}{R} + \frac{1}{R_1} + \frac{1}{R_2} \right)} \times R R_1 R_2$$

$$\Rightarrow U_{AB}'' = \frac{E_2 R R_1}{R_1 R_2 + R R_2 + R R_1}$$

$$\Rightarrow \left[ U_{AB}'' = \frac{E_2 R R_1}{R(R_1 + R_2) + R_1 R_2} \right]$$

$$\text{ii. } I'' = \frac{U_{AB}''}{R}$$

$$\Rightarrow \left[ I'' = \frac{E_2 R_1}{R(R_1 + R_2) + R_1 R_2} \right]$$

$$\kappa \quad I_1'' = -\frac{U_{AB}''}{R_1}$$

$$\Rightarrow \left[ I_1'' = \frac{-E_2 R}{R(R_1 + R_2) + R_1 R_2} \right]$$

$$\kappa \quad I_2'' = I'' - I_1''$$

$$\Rightarrow \left[ I_2'' = \frac{E_2 (R + R_1)}{R(R_1 + R_2) + R_1 R_2} \right]$$

$$\text{iii. } AN : U_{AB}'' = 2,91V ; I'' = 6,19mA$$

$$I_1'' = -2,91mA ; I_2'' = 9,09mA$$

$$\Rightarrow U_{AB} = U_{AB}' + U_{AB}'' ; AN : U_{AB} = 4,36V$$

$$I_1 = I_1' + I_1'' ; AN : I_1 = 1,64mA$$

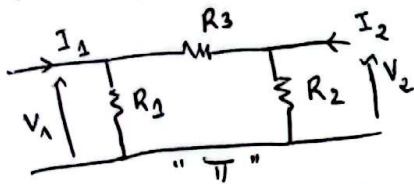
$$I_2 = I_2' + I_2'' ; AN : I_2 = 7,64mA$$

$$I = I' + I'' ; AN : I = 9,28mA$$

# TP2 : Quadripôles

## Partie théorique :

→ Manipulation 1 :



1. Matrice d'impédance Z

$$\begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases}$$

•  $Z_{11} = \frac{V_1}{I_1} \mid I_2 = 0$

$$Z_{11} = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$\Rightarrow Z_{11} = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

•  $Z_{22} = \frac{V_2}{I_2} \mid I_1 = 0$

$$Z_{22} = \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3}$$

$$\Rightarrow Z_{22} = \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3}$$

•  $Z_{12} = \frac{V_1}{I_2} \mid I_1 = 0$

$$Z_{12} = \frac{R_1}{R_1 + R_2 + R_3} V_2$$

→ Diviseur de tension :  $V_1 = \frac{R_1}{R_1 + R_3} V_2$

et  $V_2 = Z_{22} I_2 \mid I_1 = 0$

$$\Rightarrow Z_{12} = \frac{R_1}{R_1 + R_3} \cdot \frac{Z_{22} I_2}{I_2}$$

$$\Rightarrow Z_{12} = \frac{R_1}{R_1 + R_3} \cdot \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3}$$

$$\Rightarrow Z_{12} = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

•  $Z_{21} = \frac{V_2}{I_1} \mid I_2 = 0$

$$Z_{21} = \frac{R_2}{R_1 + R_2 + R_3} V_1$$

→ Diviseur de tension :  $V_2 = \frac{R_2}{R_2 + R_3} V_1$

et  $V_1 = Z_{11} I_1 \mid I_2 = 0$

$$\Rightarrow Z_{21} = \frac{R_2}{R_2 + R_3} \cdot \frac{Z_{11} I_1}{I_1}$$

$$\Rightarrow Z_{21} = \frac{R_2}{R_2 + R_3} \cdot \frac{R_1 (R_2 + R_3)}{R_1 + R_2 + R_3}$$

$$\Rightarrow Z_{21} = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$\Rightarrow Z = \begin{bmatrix} \frac{R_1 (R_2 + R_3)}{R_1 + R_2 + R_3} & \frac{R_1 R_2}{R_1 + R_2 + R_3} \\ \frac{R_1 R_2}{R_1 + R_2 + R_3} & \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3} \end{bmatrix}$$

A.N :  $R_1 = R_2 = 1 \text{ k}\Omega$   
 $R_3 = 180 \Omega$  :  $Z = \begin{bmatrix} 541,28 & 458,71 \\ 458,71 & 541,28 \end{bmatrix}$

2. Matrice de transfert T :

$$\begin{cases} V_1 = T_{11} V_2 + T_{12} (-I_2) \\ I_1 = T_{21} V_2 + T_{22} (-I_2) \end{cases}$$

•  $T_{11} = \frac{V_1}{V_2} \mid I_2 = 0$

$$T_{11} = \frac{R_2}{R_2 + R_3}$$

Diviseur de tension :  $V_1 = \frac{R_2}{R_2 + R_3} V_2$

$$\Rightarrow T_{11} = \frac{R_2}{R_2 + R_3}$$

$$\Rightarrow T_{11} = 1 + \frac{R_3}{R_2}$$

•  $T_{22} = -\frac{I_1}{I_2} \mid V_2 = 0$

$$T_{22} = -\frac{I_1}{I_2} \mid V_2 = 0$$

Diviseur de courant :  $I_1 = \frac{R_1 + R_3}{R_1} (-I_2)$

$$\Rightarrow T_{22} = -\frac{I_1}{I_2} \mid V_2 = 0 = \frac{R_1 + R_3}{R_1} \Rightarrow T_{22} = 1 + \frac{R_3}{R_1}$$

•  $T_{12} = -\frac{V_1}{I_2} \mid V_2 = 0$

$$T_{12} = -\frac{V_1}{I_2} \mid V_2 = 0$$

$$= -\frac{V_1}{I_2} \cdot \frac{I_1}{I_2} \mid V_2 = 0$$

$$= \frac{V_1}{I_1} \cdot T_{22} \mid I_1 = \frac{R_3 + R_1}{R_1} (-I_2)$$

$$\Rightarrow T_{12} = \frac{-I_2 R_3}{-I_2 \cdot \frac{(R_3 + R_1)}{R_1}} \cdot T_{22}$$

$$\Rightarrow T_{12} = \frac{R_3 R_1}{R_1 + R_3} \cdot \frac{R_1 + R_3}{R_1}$$

$$\Rightarrow T_{12} = R_3$$

$$T_{21} = \frac{I_1}{V_2} \Big|_{I_2=0} ; \text{Diagram: } R_1 \text{ in series with } R_3 \text{ in parallel with } R_2. \text{ } V_1 \text{ across } R_1, V_2 \text{ across } R_2. I_1 \text{ into } R_1, I_2 \text{ into } R_2.$$

$$= \frac{I_1}{V_1} \frac{V_1}{V_2} \Big|_{I_2=0}$$

$$T_{21} = \frac{I_1}{V_2} \cdot T_{11} ; \frac{V_1}{I_1} = Z_{in}$$

$$T_{21} = \frac{I_1}{V_2} \Big|_{I_2=0}$$

$$\Rightarrow T_{21} = \frac{T_{11}}{Z_{in}}$$

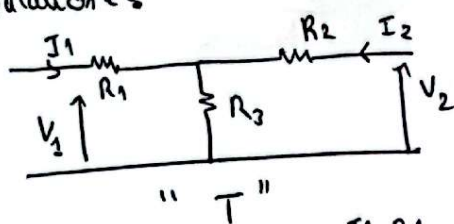
$$\Rightarrow = \frac{R_2 + R_3}{R_2} \times \frac{R_1 + R_2 + R_3}{R_2(R_2 + R_3)}$$

$$\Rightarrow \boxed{T_{21} = \frac{R_1 + R_2 + R_3}{R_1 R_2}}$$

$$T = \begin{bmatrix} 1 + \frac{R_3}{R_2} & R_3 \\ \frac{R_1 + R_2 + R_3}{R_1 R_2} & 1 + \frac{R_3}{R_1} \end{bmatrix}$$

A.N:  $T = \begin{bmatrix} 1,18 & 180 \\ 2,18 \cdot 10^{-3} & 1,18 \end{bmatrix}$

Manipulation 2:



$$1. \rightarrow Z_{E0} = \frac{V_1}{I_1} \Big|_{I_2=0} ; \text{Diagram: } R_1 \text{ in series with } R_3 \text{ in parallel with } R_2. \text{ } V_1 \text{ across } R_1, V_2 \text{ across } R_2. I_1 \text{ into } R_1, I_2 \text{ into } R_2.$$

$$\boxed{Z_{E0} = R_{eq} = R_1 + R_3} ; \text{Diagram: } R_1 \text{ in series with } R_3 \text{ in parallel with } R_2. \text{ } V_1 \text{ across } R_1, V_2 \text{ across } R_2. I_1 \text{ into } R_1, I_2 \text{ into } R_2.$$

$$0 Z_{E\infty} = \frac{V_1}{I_1} \Big|_{V_2=0} ; \text{Diagram: } R_1 \text{ in series with } R_3 \text{ in parallel with } R_2. \text{ } V_1 \text{ across } R_1, V_2 \text{ across } R_2. I_1 \text{ into } R_1, I_2 \text{ into } R_2.$$

$$\Rightarrow \boxed{Z_{E\infty} = R_1 + \frac{R_2 R_3}{R_2 + R_3}} \quad R_2 + (R_2 \parallel R_3)$$

$$2. Z_s = \frac{V_2}{I_2} \Big|_{V_1=0} ; \text{Diagram: } R_1 \text{ in series with } R_3 \text{ in parallel with } R_2. \text{ } V_1 \text{ across } R_1, V_2 \text{ across } R_2. I_1 \text{ into } R_1, I_2 \text{ into } R_2.$$

$$\boxed{Z_s = R_2 + \frac{R_1 R_3}{R_1 + R_3}} \quad (R_1 \parallel R_3) + R_2$$

3. A.N.  $Z_{E0} = 1470 \Omega \quad Z_{E\infty} = 789,72 \Omega$   
 $Z_s = 789,72 \Omega$